Maths for Computing — Phase 4: Logic and Truth Tables

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What this phase covers

This phase is all of Lecture 4. It unlocks mock **Questions 3, 5, 7, 15, 17 and 18**: six questions, a big share of your marks. It also covers revision Task 9.

Logic is the most mechanical topic in the module. If you follow the method exactly, you will get it right every time. No algebra needed, only careful table-filling.

Logic is the foundation of computer decisions: **if** statements, **Boolean** values (true/false) and logic gates in hardware all come from here.

1. Propositions

Logic is a system of rules for reasoning: deciding whether statements are true or false.

A **proposition** is a statement that is **either true or false** (never both, never neither). Propositions are named with small letters: p, q, r, s.

- "Python is a programming language." → a proposition (true).
- "Google Chrome is an antivirus." → a proposition (false).
- "2 × 2 = 5" → a proposition (false).
- "Close the door!" → **not** a proposition: an instruction can't be true or false.
- "Is it raining?" → **not** a proposition: a question.

There are only two **truth values** (also called **Boolean values**): **T (true)** and **F (false)**. Programmers write them as 1 and 0.

Atomic and compound

- An **atomic** (or **primitive**) proposition is a single simple statement: p = "I drink".
- A **compound** proposition is built from atomic ones using **logical operators** (also called **connectives**): "I drink AND I smoke" is p ∧ q.

Name	Proposition	Built from	Truth value
P1	Python is a programming language	atomic	T
P2	Google Chrome is an antivirus	atomic	F
P3	USB stands for Universal Serial Bus	atomic	T
C1	P1 AND P3	compound	T
C2	P2 OR P1	compound	T
C3	NOT P2	compound	T

A **formula** is any expression built from propositions and operators, like (p ∧ q) → p.

2. The four logical operators

Learn these four tables perfectly. Every logic question is built from them.

NOT (¬) — negation

¬p flips the truth value. Read it "not p".

p	¬p
T	F
F	T

Double negation: ¬¬p = p. Flipping twice brings you back.

AND (∧) — conjunction

p ∧ q is true **only when both** are true. Read it "p and q".

p	q	p ∧ q
T	T	**T**
T	F	F
F	T	F

p	q	p ∧ q
F	F	F

Memory aid: ∧ looks like a capital A without the bar, for **A**nd. One false makes the whole thing false.

OR (∨) — disjunction

p ∨ q is true when **at least one** is true. It's false **only when both are false**. Read it "p or q".

p	q	p ∨ q
T	T	T
T	F	T
F	T	T
F	F	**F**

Note: logical OR includes the case where both are true. "Do you want tea or coffee?" in English often means one or the other, but in logic both is fine.

IMPLIES (→ or ⇒) — implication

p → q is read "if p then q" or "p implies q". p is the **antecedent** (the "if" part); q is the **consequent** (the "then" part).

p	q	p → q
T	T	T
T	F	**F**
F	T	T
F	F	T

The one rule to remember: p → q is false only when p is true and q is false. ("True leads to false" is the only broken case.)

Why the last two rows are true — think of a promise. Your parent says: "If you pass the exam, I'll buy you a phone."

- You pass, you get the phone (T, T): promise kept. **T**.
- You pass, no phone (T, F): promise **broken**. **F**.
- You fail, you still get a phone (F, T): promise not broken; they never said what happens if you fail. **T**.
- You fail, no phone (F, F): promise not broken. **T**.

The promise can only be broken when the condition happens and the result doesn't.

Order matters: p → q is not the same as q → p. Look at row 3: p = F, q = T gives p → q = T but q → p = F.

All four in one table (copy this onto your exam paper first)

p	q	¬p	p ∧ q	p ∨ q	p → q
T	T	F	T	T	T
T	F	F	F	T	F
F	T	T	F	T	T
F	F	T	F	F	T

Mini tasks 2

Let p = T, q = F. Find:

1. ¬q
2. p ∧ q
3. p ∨ q
4. p → q
5. q → p
6. ¬p → q

3. Order of precedence and evaluating a formula

Like BIDMAS in arithmetic, logic has an order. **Precedence** means which operator is applied first.

1. **Brackets** first (innermost first)
2. ¬ NOT
3. **∧** AND
4. **∨** OR
5. **→** IMPLIES (last)

Operators of equal rank go left to right.

Memory aid: **N**ever **A**sk **O**bvious **I**mplications: Not, And, Or, Implies.

Examples

- **p ∨ q ∧ r** means p ∨ (q ∧ r), because ∧ comes before ∨. To do the OR first, you need brackets: (p ∨ q) ∧ r.
- **¬p ∧ q** means (¬p) ∧ q. The ¬ sticks only to p. Compare ¬(p ∧ q), which negates the whole bracket.
- **p ∧ q → r** means (p ∧ q) → r.

Finding the main operator

The **main operator** is the one done **last**. It decides the final column of a truth table. Find it by looking for the operator that is **outside all brackets** with the lowest precedence.

- In (p ∧ q) → p, the main operator is →.
- In ¬(p → q), the main operator is the ¬ in front (it applies to the whole bracket).
- In (¬(p ⇒ ¬q) ∨ p) ⇒ p, the main operator is the last ⇒.

Evaluating for one set of values

Replace each letter with T or F, then work from the inside out, writing each step.

Example: ¬p ∧ q → r with p = F, q = T, r = F.

1. ¬p = ¬F = T
2. ¬p ∧ q = T ∧ T = T
3. T → r = T → F = **F**

Example: p ∨ q ∧ r with p = T, q = F, r = F.

1. q ∧ r = F ∧ F = F (∧ first)
2. p ∨ F = T ∨ F = **T**

But (p ∨ q) ∧ r = (T ∨ F) ∧ F = T ∧ F = **F**. The brackets change the answer.

Mini tasks 3

1. Where do the brackets go in ¬p ∨ q ∧ r?
2. Evaluate (p → q) ∧ ¬p with p = F, q = F.
3. Evaluate ¬(p ∨ q) → r with p = F, q = F, r = F.
4. What is the main operator of (p ∨ ¬p) ⇒ (q ∧ ¬q)?

4. Building a full truth table

A **truth table** lists **every possible combination** of truth values for the atomic propositions, and the formula's value for each.

Step 1: count the rows

If the formula has **n different atomic propositions**, the table has **2^n rows**. Each letter can be T or F, so the choices double with each new letter.

Different letters	1	2	3	4	5
Rows	2	4	8	16	32

Count **different** letters only. In ¬(¬p ⇒ (q ∧ ¬q)) ∨ (¬r ⇒ (s ∨ ¬s)), the letters are p, q, r, s. q and s appear twice, but they count once. 4 letters gives **2^4 = 16 rows** (mock Q5, answer B).

Step 2: fill the letter columns with the standard pattern

The lecture's "alternating periods" method:

- The **first** letter: the top half is T, the bottom half is F.
- The **next** letter: halve the block size, alternating.
- The **last** letter: alternates every row: T, F, T, F ...

For 3 letters (8 rows):

p	q	r
T	T	T
T	T	F

p	q	r
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

p: 4 T then 4 F. q: 2 T, 2 F, 2 T, 2 F. r: T, F, T, F … This guarantees every combination appears exactly once, and matches the order your exam uses (so "Row 3" means the same thing to you and the examiner).

Step 3: add one column per sub-formula, in precedence order

Break the formula into pieces from the inside out. Each piece gets a column. The last column is the whole formula.

Step 4: fill each column using the four basic tables

Fill one column at a time, top to bottom.

Example 1 (lecture): $(p \wedge q) \rightarrow p$

2 letters → 4 rows. Columns: $p \wedge q$, then the whole.

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Row 1: T → T = T. Rows 2–4: F → anything = T. **All T.**

Example 2 (lecture): p ∧ ¬p

p	¬p	p ∧ ¬p
T	F	F
F	T	F

All F. Something can't be both true and false.

Example 3 (lecture): (p ∧ q) → (p ∨ r)

3 letters → 8 rows.

p	q	r	p ∧ q	p ∨ r	(p ∧ q) → (p ∨ r)
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	F	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	F	T	T
F	F	F	F	F	T

The final column is false only if a row has T on the left and F on the right. No row does, so **all T**.

Example 4: (p ∨ ¬q) → ¬p

p	q	¬q	p ∨ ¬q	¬p	(p ∨ ¬q) → ¬p
T	T	F	T	F	**F**
T	F	T	T	F	**F**
F	T	F	F	T	T
F	F	T	T	T	T

Final column: F, F, T, T. "FALSE" appears **2 times**.

Speed tip for → columns

For any implication column, scan for rows where the **left is T and the right is F**. Those rows are F; **every other row is T**. That's much faster than checking every row.

Mini tasks 4

1. How many rows does a truth table for (p → q) ∨ (r ∧ ¬p) have?
2. Build the full truth table for ¬p ∨ q. Compare it with p → q. What do you notice?
3. Build the truth table for p → (q ∧ r). How many F values are in the final column?

5. Valid, satisfiable and unsatisfiable formulas

Once the final column is filled in, classify the formula by looking at it:

Final column	Name	Meaning	Example
All T	**Valid** (also called a **tautology**)	Always true, whatever the values	p ∨ ¬p; (p ∧ q) → p
At least one T	**Satisfiable**	True for some values	p → q; p ∧ q
No T at all (all F)	**Unsatisfiable** (also called a **contradiction**)	Never true	p ∧ ¬p

How they relate:

- **Every valid formula is also satisfiable** (all T certainly includes at least one T).
- A formula can be **satisfiable but not valid**: some T, some F. Example: p → q.
- **Unsatisfiable** is the opposite of satisfiable. And a formula is valid exactly when its negation is unsatisfiable.

Example (mock Q3): (p ∨ ¬p) ⇒ (q ∧ ¬q). The left side, p ∨ ¬p, is always T. The right side, q ∧ ¬q, is always F. So every row is T ⇒ F = F. The formula is unsatisfiable, and FALSE appears in **all 4 rows**.

Mini tasks 5

Classify each as valid, satisfiable but not valid, or unsatisfiable:

1. p → p

2. p ∧ q
3. (p → q) ∧ p ∧ ¬q
4. ¬(p ∧ ¬p)
5. p ∨ q

6. Logical consequence and knowledge bases

The idea

A **knowledge base (KB)** is a set of statements we accept as true, our known facts. We often want to know: if all of these are true, **must** some other statement C also be true?

If it must, C is a **logical consequence** of the KB. (We also say C **follows from** the KB.)

The test (memorise this)

C is a logical consequence of A_1, A_2, ..., A_n exactly when this formula is **valid** (all T):

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(A_1 \wedge A_2 \wedge \dots \wedge A_n) \Rightarrow C
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In words: join every KB statement with AND on the left, put C on the right, connect with an arrow. Build the truth table. **All T in the final column → C follows. Any F → C does not follow.**

Why this works

The implication is only false when the left is T and the right is F. That would be a situation where **all the facts are true but C is false**, which proves C doesn't have to follow. If there's no such row, C is guaranteed.

Fast method: you don't always need the full table. Look only at rows where **every KB statement is T**. If C is T in all of those rows, C follows. If C is F in any of them, it doesn't.

The Good Tenants problem (lecture)

A landlady wants a tenant who does not drink and does not smoke. Let p = "I drink", q = "I smoke".

- **Student A:** "If I drink, then I smoke, but I do not drink." KB: p ⇒ q, ¬p. Does it follow that A doesn't smoke (¬q)?

- **Student B:** "If I drink, then I smoke, but I do not smoke." KB: $p \Rightarrow q$, $\neg q$. Does it follow that B doesn't drink ($\neg p$)?

Student A: check $((p \Rightarrow q) \wedge \neg p) \Rightarrow \neg q$.

p	q	¬p	p ⇒ q	(p ⇒ q) ∧ ¬p	¬q	final
T	T	F	T	F	F	T
T	F	F	F	F	T	T
F	T	T	T	T	F	**F**
F	F	T	T	T	T	T

Row 3 is F: A might not drink but still smoke. **Not valid, so ¬q does not follow.** We can't be sure A doesn't smoke.

Student B: check $((p \Rightarrow q) \wedge \neg q) \Rightarrow \neg p$.

p	q	p ⇒ q	¬q	(p ⇒ q) ∧ ¬q	¬p	final
T	T	T	F	F	F	T
T	F	F	T	F	F	T
F	T	T	F	F	T	T
F	F	T	T	T	T	T

All T: valid. B certainly doesn't smoke (they said so) and certainly doesn't drink (if they did, they'd smoke). **B gets the room.**

Revision Task 9c: KB = $\{p \vee \neg q, q \rightarrow p\}$. Does p follow?

p	q	¬q	p ∨ ¬q	q → p	KB (both)	KB → p
T	T	F	T	T	T	T
T	F	T	T	T	T	T
F	T	F	F	F	F	T
F	F	T	T	T	T	**F**

In the last row both KB facts are true but p is false. **p is not a logical consequence.**

Revision Task 9b: KB = {q ∧ ¬r}. Does r follow?

The KB is true only when q = T, r = F. In that row r is F, so (q ∧ ¬r) → r is F there. **r does not follow.** (In fact the KB tells us r is false.)

Revision Task 9a: a contradictory KB

KB = {¬q ∧ q}. This is never true. So (¬q ∧ q) → anything has a left side that is always F, which makes every row T. **Valid.**

This is a strange but important rule: **from a contradiction, anything follows.** Since the KB can never be true, there is never a row with true facts and a false conclusion.

Mini tasks 6

Use the test or the fast method:

1. KB = {p → q, p}. Does q follow?
2. KB = {p → q, q}. Does p follow?
3. KB = {p ∨ q, ¬p}. Does q follow?
4. KB = {p ∨ q}. Does p follow?

7. Shortcuts that save time

These aren't in your lecture slides, but they're true, and they let you check answers fast. If unsure, you can always fall back on a full truth table.

Equivalences

Two formulas are **equivalent** (≡) when their truth tables have the same final column.

Formula	Equivalent to	Why it helps
¬¬p	p	Two NOTs cancel
p → q	¬p ∨ q	Turns an arrow into an OR
¬(p → q)	p ∧ ¬q	True only when p is T and q is F
p → q	¬q → ¬p	The **contrapositive**: always the same truth value
¬(p ∧ q)	¬p ∨ ¬q	**De Morgan's law**: NOT flips AND into OR

Formula	Equivalent to	Why it helps
¬(p ∨ q)	¬p ∧ ¬q	**De Morgan's law**: NOT flips OR into AND

And two facts: **p ∨ ¬p is always T**, and **p ∧ ¬p is always F**.

Set connection: these match the set laws from Phase 2. ∧ behaves like ∩, ∨ like ∪, ¬ like the complement. For example, (A ∪ B)′ = A′ ∩ B′ is De Morgan's law for sets.

Common argument patterns

KB	Conclusion	Follows?	Name
p → q, p	q	**Yes**	Modus ponens
p → q, ¬q	¬p	**Yes**	Modus tollens (Student B)
p → q, q → r	p → r	**Yes**	Chain rule (hypothetical syllogism)
p ∨ q, ¬p	q	**Yes**	Disjunctive syllogism
p → q, ¬p	¬q	**No**	Fallacy: denying the antecedent (Student A)
p → q, q	p	**No**	Fallacy: affirming the consequent
Any contradiction	anything	**Yes**	From a contradiction, anything follows

Quick test for "when is a formula false?"

Work backwards from the main operator:

- **A → B is false** only when A = T and B = F.
- **A ∨ B is false** only when both are F.
- **A ∧ B is false** when either is F.
- **¬A is false** when A is T.

8. Your mock logic questions, fully solved

Q3: How many FALSE values are in the final column of $(p \vee \neg p) \Rightarrow (q \wedge \neg q)$?

p	q	¬p	p ∨ ¬p	¬q	q ∧ ¬q	final
T	T	F	T	F	F	F
T	F	F	T	T	F	F
F	T	T	T	F	F	F
F	F	T	T	T	F	F

T ⇒ F in every row. **4 times. Answer B.**

Q5: How many rows for $\neg(\neg p \Rightarrow (q \wedge \neg q)) \vee (\neg r \Rightarrow (s \vee \neg s))$?

Different letters: p, q, r, s. That's 4, so 2^4 = **16 rows. Answer B.** (8 is the trap for counting only 3 letters.)

Q7: Which values make $\neg(p \Rightarrow q)$ true?

¬(p ⇒ q) is true exactly when p ⇒ q is false. That happens only when p = T and q = F. **Answer C.**

Q15: Spot the wrong row in the truth table for $(\neg(p \Rightarrow \neg q) \vee p) \Rightarrow p$

Recompute every column yourself and compare with the given table:

Row	p	q	¬q	p ⇒ ¬q	¬(p ⇒ ¬q)	¬(p ⇒ ¬q) ∨ p	final
1	T	T	F	F	T	T	T
2	T	F	T	T	F	T	T
3	F	T	F	T	F	**F**	T
4	F	F	T	T	F	F	T

In row 3: ¬(p ⇒ ¬q) = F and p = F, so F ∨ F = **F**. The given table says **T** there. **The mistake is in Row 3. Answer C.**

(Notice the final column is all T anyway. The error is hidden in a middle column, so you must check every column, not just the last.)

Q17: In which case is C a logical consequence of KB?

Use the fast method: find a row where every KB fact is T but C is F. If one exists, C doesn't follow.

- **(A)** KB {p ⇒ ¬q, ¬q}, C = p ∧ q. Try p = F, q = F: p ⇒ ¬q = F ⇒ T = T, ¬q = T. Both facts true, but C = F ∧ F = F. **Doesn't follow.** (Even simpler: ¬q says q is false, so p ∧ q can't be true.)
- **(B)** KB {p ⇒ q, q ⇒ r}, C = p ⇒ r. This is the chain rule. Full check:

p	q	r	p ⇒ q	q ⇒ r	KB	p ⇒ r	KB ⇒ C
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

All T. **C follows. Answer B.**

- **(C)** KB {¬p ⇒ q, ¬p}, C = p ∧ q. ¬p means p = F, so p ∧ q is F. **Doesn't follow.**
- **(D)** KB {p ⇒ q, q ∧ ¬p}, C = p. q ∧ ¬p means p = F, so C is F. **Doesn't follow.**

Q18: For which inputs is ¬¬((p ⇒ q) ∨ (q ⇒ p)) FALSE?

¬¬ cancels, leaving (p ⇒ q) ∨ (q ⇒ p).

p	q	p ⇒ q	q ⇒ p	OR
T	T	T	T	T
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Never false (it's valid). **Answer A: none of the inputs give FALSE.**

Phase 4 practice questions (exam style)

About 30 minutes. Copy the four basic tables onto your paper first, just as you will in the exam.

1. How many rows are in the truth table for ((p ∧ q) → (r ∨ ¬s)) ∨ p? (A) 8 (B) 16 (C) 32 (D) 5
2. When is (p ∨ q) ∧ ¬p true? (A) p T, q T (B) p T, q F (C) p F, q T (D) p F, q F
3. How many times does FALSE appear in the final column of (p ∨ q) → r? (A) 1 (B) 2 (C) 3 (D) 5
4. Which formula is valid? (A) p → q (B) p ∨ ¬p (C) p ∧ ¬p (D) p ∨ q
5. Which formula is unsatisfiable? (A) p → ¬p (B) ¬(p ∨ ¬p) (C) p ∧ q (D) ¬p ∨ q
6. The only values that make ¬(p ∨ q) true are: (A) p T, q T (B) p T, q F (C) p F, q T (D) p F, q F
7. Evaluate ¬p ∧ q → r with p = F, q = T, r = F.
8. Evaluate p ∨ q ∧ ¬r with p = F, q = T, r = T.
9. KB = {p → q, p}. Is q a logical consequence?
10. KB = {p → q, q}. Is p a logical consequence?
11. KB = {¬q → ¬p, p}. Is q a logical consequence?
12. KB = {p ∨ q, ¬q}. Is p a logical consequence?
13. Which is equivalent to ¬(p ∧ q)? (A) ¬p ∧ ¬q (B) ¬p ∨ ¬q (C) p ∨ q (D) ¬p ∨ q
14. Which is equivalent to p → q? (A) q → p (B) ¬p → ¬q (C) ¬q → ¬p (D) p ∧ ¬q
15. A student built this truth table for (p → q) ∧ ¬p. Which row contains a mistake?

Row	p	q	p → q	¬p	(p → q) ∧ ¬p
1	T	T	T	F	F
2	T	F	T	F	F
3	F	T	T	T	T
4	F	F	T	T	T

16. How many times does TRUE appear in the final column of (p ∧ q) ∨ r? (A) 3 (B) 4 (C) 5 (D) 7
17. "If it rains, the ground is wet. The ground is not wet. So it did not rain." Is this conclusion a logical consequence? Name the pattern.
18. For which values is ¬(p → q) ∨ q FALSE? (A) p T, q F (B) p F, q F (C) p T, q T (D) never

19. How many rows does a truth table with 5 different propositions have?
20. Classify (p → q) ∨ (q → p): valid, satisfiable but not valid, or unsatisfiable?

Answers with full explanations

Mini tasks 2 (p = T, q = F)

1. ¬q = **T**
2. T ∧ F = **F**
3. T ∨ F = **T**
4. T → F = **F** (the only broken case)
5. q → p = F → T = **T**
6. ¬p → q = F → F = **T**

Mini tasks 3

1. **(¬p) ∨ (q ∧ r)**: ¬ first, then ∧, then ∨.
2. p → q = F → F = T. ¬p = T. T ∧ T = **T**.
3. p ∨ q = F, so ¬(p ∨ q) = T. T → F = **F**.
4. The ⇒ in the middle (outside all brackets).

Mini tasks 4

1. Letters p, q, r: **8 rows**.
2. ¬p ∨ q: TT → F ∨ T = T; TF → F ∨ F = F; FT → T; FF → T. Final column **T, F, T, T**, identical to p → q. **They are equivalent.**
3. p → (q ∧ r) is F when p = T and q ∧ r = F. With p = T, q ∧ r is F in rows TTF, TFT and TFF. **3 F values.**

Mini tasks 5

1. **Valid.** T → T and F → F are both T.
2. **Satisfiable but not valid.** Only T in the first row.
3. **Unsatisfiable.** It needs p = T and q = F, but then p → q = F.
4. **Valid.** p ∧ ¬p is always F, so its negation is always T.
5. **Satisfiable but not valid.** F only when both are F.

Mini tasks 6

1. **Yes** (modus ponens). The only row with both facts true is p = T, q = T.
2. **No.** p = F, q = T makes both facts true but p false.
3. **Yes.** ¬p makes p false, so for p ∨ q to hold, q must be true.
4. **No.** p = F, q = T makes p ∨ q true but p false.

Phase 4 practice questions

1. **(B)** Different letters: p, q, r, s. $2^4 = 16$.
2. **(C)** ¬p needs p = F. Then p ∨ q needs q = T.
3. **(C)** F when p ∨ q = T and r = F: rows TTF, TFF and FTF. Three rows.
4. **(B)** Something is always either true or not true.
5. **(B)** p ∨ ¬p is always T, so its negation is always F. (A) p → ¬p is T when p = F.
6. **(D)** ¬(p ∨ q) is true when p ∨ q is false, which needs both false. (De Morgan: ¬p ∧ ¬q.)
7. **F.** ¬p = T; T ∧ T = T; T → F = F.
8. **F.** ¬r = F; q ∧ F = F (∧ before ∨); F ∨ F = F.
9. **Yes.** Modus ponens.
10. **No.** Affirming the consequent. Counterexample: p = F, q = T.
11. **Yes.** ¬q → ¬p is the contrapositive of p → q, so this is modus ponens again. Check: p = T forces ¬p = F; for ¬q → ¬p to be true, ¬q must be F, so q = T.
12. **Yes.** Disjunctive syllogism: q is false, so p must be true.
13. **(B)** De Morgan's law.
14. **(C)** The contrapositive. (A) and (B) are common traps: they are not equivalent. (D) is the negation.
15. **Row 2.** With p = T, q = F, p → q is **F**, not T. (The final value F happens to be right anyway, since ¬p = F. That's why you must check every column.)
16. **(C)** r = T gives 4 rows (all T). With r = F, only p = T, q = T gives T: 1 more. 4 + 1 = 5.
17. **Yes, modus tollens.** p = "it rains", q = "the ground is wet". KB = {p → q, ¬q}, conclusion ¬p. Same as Student B.
18. **(B)** ¬(p → q) ∨ q is false when both parts are false: q = F, and ¬(p → q) = F, meaning p → q = T. With q = F, p → q = T needs p = F. So p = F, q = F. Check: ¬(F → F) ∨ F = ¬T ∨ F = F. ✓
19. 2^5 = **32 rows**.

20. **Valid.** If q = T, then p → q = T. If q = F, then q → p = T. Either way, one side is true. (Same as mock Q18.)

Scoring: 17 or more out of 20 means you're ready. Logic rewards care, not cleverness: always write every column.

When you're ready, ask for **Phase 5: the full mock solved and a harder practice exam**.